
When Shift Happens - Confounding Is to Blame

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Abstract

1 Distribution shifts introduce a form of epistemic and statistical uncertainty that
2 can substantially erode the robustness and generalization capacity of contemporary
3 machine learning models. Although the prevailing causal perspective maintains
4 that robustness should be promoted by learning causal-invariant representations,
5 recent empirical studies have produced a strikingly counterintuitive pattern: (i)
6 empirical risk minimization (ERM) can match, and in several settings even surpass,
7 state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) its OOD
8 performance improves when prediction exploits all available covariates rather
9 than only those designated as causal. On the basis of complementary theoretical
10 and empirical evidence, we argue that this phenomenon is best understood as a
11 consequence of hidden confounding. Shifts in unobserved confounders induce
12 distributional changes that violate assumptions routinely imposed by existing
13 OOD generalization procedures; under these conditions, we prove that successful
14 generalization requires the recovery of environment-specific relationships rather
15 than exclusive reliance on invariant ones. We further show that augmenting models
16 with proxy variables for hidden confounders can attenuate the difficulties introduced
17 by hidden confounding shifts. Collectively, these results provide both theoretical
18 clarification and practical guidance for the design of robust OOD generalization
19 algorithms and principled covariate-selection strategies.

20 1 Introduction

21 Generalization—the capacity to infer reliable conclusions about unobserved data from finite observa-
22 tions—occupies a foundational position across a wide range of scientific disciplines. In medicine and
23 the social sciences, this concern appears as *external validity*, where conclusions obtained in one popu-
24 lation must remain applicable to another [13]; in ecology, it underwrites *space-for-time substitutions*,
25 through which spatial heterogeneity is mobilized as evidence about temporal ecological change [60];
26 and in engineering, it is formalized through *robust control*, where model performance must persist
27 despite unmodeled disturbances [35]. Within this broader intellectual lineage, machine learning has
28 increasingly become a central instrument for constructing models whose learned regularities are
29 intended to transfer beyond the data on which they were estimated [100, 88, 49].

30 For machine learning models to generalize reliably, they must retain predictive adequacy when
31 the training and test distributions diverge. For instance, a model trained to predict *food stamp*
32 *reciprocity* from *household attributes* in one state should continue to perform effectively when
33 transferred to another state whose institutional and demographic profile may differ. A model is
34 said to exhibit *out-of-distribution (OOD) generalization* when it sustains performance not only
35 on in-distribution (ID) data, from which the training sample is drawn, but also on OOD test data
36 generated under shifted conditions. During the past decade, methodological families including
37 domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and
38 robust learning [42, 69, 94] have been developed to provide provable OOD generalization under
39 specified assumptions. Yet, when model selection is performed carefully [30, 38, 87], standard
40 *empirical risk minimization (ERM)* [86] frequently attains competitive OOD performance across
41 diverse real-world applications [30, 24, 50, 57, 67, 87]. Moreover, a recent empirical analysis of 16
42 real-world tabular datasets [57] reports that prediction can improve when all available covariates are

incorporated, even when some of those variables are not direct causes of the outcome. Together, these observations unsettle standard intuitions in the field and motivate a more careful examination of the mechanisms that govern OOD generalization.

Distribution shifts naturally arise when observations are generated in heterogeneous environments [71]. In the food-stamp example, *food stamp reciprocity* may vary across states because eligibility rules, administrative practices, and population characteristics induce distinct environment-specific distributions. Within such data, some statistical relationships between covariates and outcomes may remain stable across environments; these are typically described as *invariant relationships* [6]. The identification and exploitation of invariances has therefore become a central route to OOD generalization [6, 59, 31, 66, 56, 98, 63]. A particularly influential instance is *causal invariance*, in which the relevant causal mechanisms are assumed to remain stable across environments [71, 59, 6]. In many real-world settings, however, the full covariate set required for accurate prediction is unavailable because of data-collection constraints, privacy restrictions, or measurement error [14, 52, 20, 48]. The resulting omissions can disrupt the very invariances on which generalization methods depend, especially when the missing variables are confounders that jointly influence observed covariates and outcomes. Because such hidden confounders are pervasive, and because shifts in their distributions instantiate distinct environments, ignoring them can both degrade generalization performance [40, 4, 85, 62] and induce models to learn erroneous relationships between observed covariates and the outcome.

Despite this importance, the consequences of hidden confounding shifts for OOD generalization remain comparatively underexamined [4, 40, 85, 62]. The present work develops theoretical arguments and empirical evidence that explain several recent observations in the OOD literature [30, 24, 50, 57, 67, 87]. Our principal contributions are as follows:

- We motivate, from a causal perspective, the systematic study of confounding shift in OOD generalization (§ 3) and explain why informative non-causal covariates can improve predictive performance.
- We show that maximizing predictive information between model predictions and true outcomes requires the explicit recovery of environment-specific relationships; consequently, invariant representations, although necessary, are not by themselves sufficient (§ 4.1, § 4.2).
- We demonstrate that variables informative about either the outcome or hidden confounders can increase the predictive information shared by model predictions and true outcomes, thereby emphasizing the importance of principled covariate selection under hidden confounding shift (§ 4.3).
- Through experiments on real-world and synthetic datasets, we provide evidence that (i) hidden confounding is prevalent in real-world tabular benchmarks, (ii) learning environment-specific relationships is positively associated with ID and OOD test accuracy, and (iii) incorporating informative non-causal covariates improves generalization (§ 5).

Overall, the analysis explains why ERM can appear unexpectedly effective under real-world distribution shifts while simultaneously foregrounding the subtle, persistent, and often unavoidable role of hidden confounding in OOD generalization, thereby motivating methods that confront this source of instability more directly.

2 Related work

Out-of-distribution (OOD) generalization comprises several partially overlapping research programs, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Whereas domain adaptation presumes access to unlabeled test-domain data, domain generalization typically relies on environment annotations during training; federated learning formulates multi-environment learning as a collaborative procedure constrained by communication and privacy, and OOD detection seeks to identify samples that depart substantially from the training distribution. Although many of these approaches share an emphasis on invariant relationship learning, recent work indicates that invariance alone does not supply a sufficiently rich inductive bias for strong generalization [46, 72, 93]. In particular, Ye et al. [93] argue that feature invariance is necessary but insufficient and that the *informativeness* of learned representations must also be accounted for. Our analysis likewise treats invariance and informativeness as jointly consequential dimensions of OOD generalization.

Proxy variable adjustment: When confounding variables are available during training but absent at test time, Landeiro and Culotta [40] propose explicit adjustment for confounding shifts as a means of improving classifier performance. Extending proxy-adjustment strategies from causal effect estimation [55, 39], recent domain-adaptation work has recast unknown distribution shift as a causal-identification problem [4, 85]. OOD generalization under hidden confounding shift has also recently been studied under the assumption that confounder support overlaps across training and test environments [62]. In contrast to these approaches, our objective is explanatory: we characterize *how* proxy variables improve OOD generalization and which information-theoretic quantities mediate that improvement.

All variable models vs causal models. Recently, Nastl and Hardt [57] introduced a benchmark in which covariates are organized into four categories: conservatively chosen causal variables, arguably causal variables, anti-causal variables, and other spurious covariates. Across 16 benchmark datasets, they show that models using all covariates Pareto-dominate models restricted to causal or arguably causal subsets on both ID and OOD data, yet the theoretical basis for this pattern remains limited. We provide scenarios and arguments that explain these experimental findings. For linear causal models, anchor regression [68] offers a framework that interpolates between two estimation paradigms: models that use all observed covariates and models restricted to causal covariates. Our concern is complementary, namely to explain why adding covariates that are not necessarily causal can be beneficial under hidden confounding shift. Relatedly, Eastwood et al. [21] show that unstable covariates can improve performance when they contain label information and are conditionally independent of stable covariates given the label; their method adjusts distribution shift by inspecting the unlabeled test domain. Nevertheless, when applied to a medical real-world dataset not constructed for that specific problem [8], ERM remains competitive with their approach, consistent with the findings of Nastl and Hardt [57]. This pattern accords with the broader observation that, under well-specified covariate shifts, maximum likelihood estimation (MLE) is minimax optimal for OOD generalization [26]. Real-world settings, however, are rarely so well specified, precisely because hidden confounding shift can disturb the assumptions on which such guarantees depend; this complication is the main focus of our work.

3 Manifestations of hidden confounding shift

This section supplies the causal and distributional background needed to formalize hidden confounding shift, and it motivates the need to address such shifts through an example concerning the identification of causal effects from observed covariates to the outcome.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , one may observe several distribution shifts between two environments e and e' as shown in Table 1. These distribution shifts usually result from a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes either $\mathbf{X}$ or Y or both (see Figure 1). The shifts in $\mathbb{P}(U)$ lead to shifts in observed distributions involving only $\mathbf{X}$ and Y . For instance, we can write: $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U) \mathbb{P}(\mathbf{X} | U)$. Thus, when $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, we observe $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. Similar arguments can be made about the distribution shifts of $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

Table 1: Taxonomy of distribution shifts induced by environment-specific changes in observable or latent distributions.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

Several existing methods for OOD generalization assume certain causal relationships among U , $\mathbf{X}$, Y that guarantee specific invariances. For instance, when $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y) \mathbb{P}(\mathbf{X} | Y)$, a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, but conditional covariate distribution stays invariant i.e., $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to covariate shift, i.e., $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, but the conditional distribution $\mathbb{P}(Y | \mathbf{X})$ stays invariant i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These

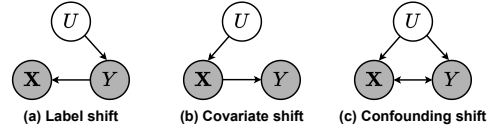


Figure 1: Causal graph structures that give rise to the corresponding distribution-shift regimes.

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

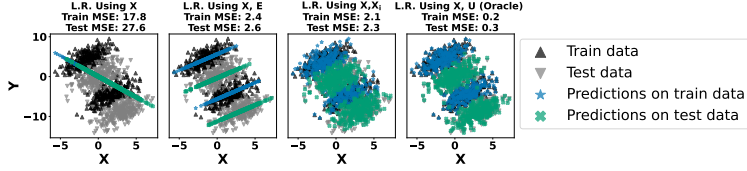


Figure 2: Four linear regression (L.R.) models are evaluated in an OOD setting with hidden confounding shifts and minimal environment overlap (i.e., distant μ_e). (i) A model trained only on X learns an incorrect relationship with Y , illustrating Simpson’s paradox. (ii) Incorporating environment-specific summary statistics of X , denoted by E , recovers the correct relationship but remains representationally limited. (iii) Adding an informative covariate X_i for U improves OOD generalization. (iv) The oracle model is trained on X and U .

invariances may not hold in many scenarios because U usually causes both X and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In this case, when $\mathbb{P}(U)$ shifts between environments, label shift, covariate shift, conditional covariate shift, and concept shift can all occur simultaneously. Nevertheless, confounding shift induces an invariance: $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ referred to as *stable confounding shift* [85, 4]. Since U is unobserved, this invariance is absent from the observed data.

Information-theoretic measures: Following Federici et al. [22], we quantify distribution shifts using mutual information. Specifically, with E denoting the environment variable, we use $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$ to measure, respectively, cross-environmental shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$. Existing OOD generalization methods often implicitly require either $I(\mathbf{X}; E | Y) = 0$, as in label-shift settings, or $I(Y; E | \mathbf{X}) = 0$, as in covariate-shift settings; under hidden confounding shift, however, both quantities can remain nonzero.

Challenges with hidden confounding shift: Figure 2 illustrates the difficulty that hidden confounding shift creates for OOD generalization. The outcome Y depends on both X and an unobserved confounder U , and environment-specific changes in the mean of U , denoted by μ_e , generate distinct environments. A linear regression model trained only on the observed covariate X estimates the wrong relationship between X and Y , as shown in the first subplot; this is an instance of Simpson’s paradox [73], because Y increases with X within each environment even though the pooled model learns the opposite trend. Similar behavior appears for OOD generalization methods such as IRM [6], VREX [38], and Group DRO [69]. In such settings, our theoretical analysis in § 4 shows that the optimal generalization strategy requires learning environment-specific relationships, which permits recovery of the correct X – Y relationship. The second subplot demonstrates that environment-specific summary statistics of X , including means, standard deviations, and quantiles, can help uncover this relationship. This behavior parallels the *backdoor adjustment* criterion in causal effect estimation [58], with environment-specific information functioning as a proxy for confounder adjustment. Nevertheless, because such information encoded solely in observed covariates can have limited representational capacity, additional informative covariates that proxy unobserved confounders can further improve generalization, as shown in the third subplot and analyzed in § 4.3. For comparison, the oracle model is trained on both X and U and therefore attains the best-possible mean squared error.

4 Impact of hidden confounding shifts on OOD generalization

This section provides a theoretical analysis of how hidden confounding shifts alter several central aspects of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the predicted label for input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. We write $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ for entropy and $I(X; Y) = \mathbb{E}_{X, Y}[\log \frac{\mathbb{P}(X, Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ for mutual information. The risk of predictor $(f \circ \phi)$ in environment e is $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$, where $\ell(\cdot, \cdot)$ is a loss function. OOD generalization seeks a predictor $(f \circ \phi)$ that maintains performance on both ID and OOD data, and the literature has consequently proposed several objective families. Robust optimization methods [7, 9, 19, 75, 69] minimize worst-environment risk, $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, with $\mathcal{E}_{tr}$ denoting the training environments. Variance risk extrapolation (VREX) [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, where β is a hyperparameter. Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to the constraint that f is simultaneously optimal across all environments. By contrast, empirical risk minimization (ERM) pools data from all training environments and minimizes empirical risk on the

aggregate sample [6, 38], namely $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ denotes observed data in environment e .

To clarify how hidden confounding shift interacts with these objectives, we analyze the maximization of predictive information, $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. Because $I(Y; \hat{Y})$ measures the information that the prediction $\hat{Y}$ carries about the true label Y , it provides a natural objective for many supervised learning problems. We begin by formalizing two representation-level properties of $\phi(\mathbf{X})$ —*informativeness* and *invariance*—that are both central to OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y | E)$.*

Conditional informativeness measures how much information $\phi(\mathbf{X})$ retains about Y within each environment. Minimizing CINF therefore entails a loss of predictive information, although maximizing it can be undesirable in settings such as algorithmic fairness, where $\phi(\mathbf{X})$ may exploit sensitive or biased within-environment signals and thereby generate unfair predictions. Although such biased representations can be discouraged by minimizing $I(\phi(\mathbf{X}); E)$, doing so may reduce predictive performance, as we show in § 4.2. In the setting where E is not associated with sensitive information, we therefore adopt and subsequently substantiate the view that maximizing CINF can improve generalization performance § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E | Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E | Y)$.*

Invariance quantifies the degree to which $\phi(\mathbf{X})$ remains consistent across environments after conditioning on a given label Y . Although reduced invariance can promote overfitting by retaining environment-specific information in $\phi(\mathbf{X})$, increased invariance suppresses these dependencies and thereby encourages invariant learning. Extending the information-theoretic measures introduced in § 3, we use $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to quantify *feature shift* and *concept shift*, respectively. Depending on context, *concept shift* refers either to $I(Y; E | \mathbf{X})$ or to $I(Y; E | \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

We begin with a few causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are marginally independent but become conditionally dependent when conditioning on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$. If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$. A path is open if it is not blocked.

Now consider the predictive information $I(Y; \hat{Y})$, where the predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are based on a learned representations $\phi(\mathbf{X})$. We model the underlying causal relationships among $\mathbf{X}, Y, U, \phi(\mathbf{X})$, $\hat{Y}$ and E as shown in Figure 3. Here, U is a hidden confounding variable. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$, $\hat{Y}$ is redundant for reasoning about Y . That is, $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. E denotes an environment variable that captures shifts in $\mathbb{P}(U)$. That is, between any two environments, there is a shift in $\mathbb{P}(U)$. We first present a more general decom-

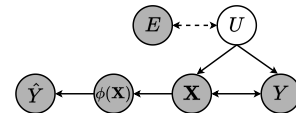


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y indicates that some covariates in $\mathbf{X}$ may cause Y , whereas others may themselves be caused by Y .

position of predictive information without explicit consideration of how U influences $\mathbf{X}, Y$. A formal treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are provided in Appendix § A. Existing methods can be understood as procedures that deliberately optimize particular components of Equation (1). For example, IRM [6] requires a fixed classifier f to remain optimal across environments, a requirement directly connected to maximizing $-I(\phi(\mathbf{X}); E|Y)$: if $\phi(\mathbf{X})$ retained additional environment-specific information conditional on Y , the same classifier would generally cease to be optimal in every environment. DANN [23] and the independence criterion for fair classification [22] seek to remove environment-specific information from $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$, and domain-adaptation methods often learn feature properties that are invariant to domains [82, 90]. Although minimizing $I(\phi(\mathbf{X}); E)$ may be desirable for fair classification, it can also impair predictive performance [98, 33, 22], as Equation (1) shows that reducing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to enforce $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$; this equality renders E independent of $(\phi(\mathbf{X}), Y)$ and consequently implies $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, even though CDAN does not explicitly penalize $I(Y; E | \phi(\mathbf{X}))$, its joint-distribution alignment objective effectively drives this conditional mutual information toward zero. Finally, while $I(Y; E)$ is constant for a fixed data-generating process, several methods explicitly seek robustness to shifts in label distributions [69, 95].

Equation (1) was derived without specifying how U influences $\mathbf{X}$ and Y , but this causal structure can determine which terms should be controlled. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, making the minimization of $I(\phi(\mathbf{X}); E | Y)$ a principled objective. Similarly, if $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, motivating minimization of $I(Y; E | \phi(\mathbf{X}))$. When $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, however, the terms in Equation (1) interact in a more intricate way, leaving open the question of what an ideal strategy should optimize. The next section resolves this question.

4.2 Predictive information under hidden confounding shift

To determine how a hidden confounder U that causes both $\mathbf{X}$ and Y modifies predictive information, we analyze *two special cases* for the relationship between $\mathbf{X}$ and Y : (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider Y , thereby allowing additional information to flow from $\phi(\mathbf{X})$ to E through Y . Conversely, conditioning on $\phi(\mathbf{X})$ partially blocks information flow from Y to E at $\mathbf{X}$ whenever $\phi(\mathbf{X})$ encodes information about $\mathbf{X}$.

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \quad \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Analogously, when $Y \rightarrow \mathbf{X}$, the inequalities in (3) arise because conditioning on Y blocks the flow of information from $\phi(\mathbf{X})$ to E at Y , whereas conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at $\mathbf{X}$. This opening occurs because $\phi(\mathbf{X})$ is a child of $\mathbf{X}$, and conditioning on a child of a collider activates the path through that collider [58], as recalled in the preliminaries of § 4.1.

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \quad \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2), (3), we can therefore refine the predictive-information decomposition as follows.

Proposition 4.2. For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if

either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

Proposition 4.2 treats the two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, although the former is more prevalent in real-world tabular prediction tasks. For example, 11 of the 16 datasets examined in recent benchmark studies [57, 50, 24] follow the causal structure $\mathbf{X} \rightarrow Y$; in the remaining 5 datasets, covariates caused by Y are far fewer than covariates that cause Y . The principal implication of Proposition 4.2 is summarized below.

Relative to the decomposition in (1), the decomposition in (4)—obtained for the special cases (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ —eliminates the terms *variation, label shift, feature shift, and concept shift*. Thus, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$ holds, predictive information is maximized by maximizing the difference *conditional informativeness* – *residual*.

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Maximizing conditional informativeness: In recent work, Prashant et al. [62] propose a mixture-of-experts (MoE) model for OOD generalization under hidden confounding shift, with each expert associated with a particular hidden-confounder assignment. Each expert therefore specializes in maximizing performance within the environment corresponding to a given confounder value, which aligns MoE learning with maximization of conditional informativeness $I(\phi(\mathbf{X}); Y|E)$. Equation (4) supplies theoretical support for MoE-type models, while also indicating the need for approaches that remain valid in more general confounding-shift regimes. For example, the method of Prashant et al. [62] assumes overlapping confounder support, i.e., $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, and discrete-valued proxy variables.

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The results in Figure 2 demonstrate that, even in a simple setting without confounder-support overlap and with a continuous-valued proxy, a linear regression model supplied with environment-specific information can recover the correct causal relationship between observed covariates and the target across environments. This example motivates OOD generalization methods that remain effective when confounder support is limited and the confounder is continuous-valued.

Minimizing residual: The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies how much additional information $\phi(\mathbf{X})$ provides about the true label Y beyond what is already contained in the prediction $\hat{Y}$. The residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ is related to the expected cross-entropy loss as: $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$ is known as the *refinement error* that measures the model’s ability to distinguish between classes. Minimizing the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus involves two complementary strategies. First, one can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can minimize the refinement error $H(Y | \hat{Y})$, for instance, through training techniques that minimize refinement error for accurate predictions before minimizing the calibration error [10].

4.3 Impact of additional informative variables

In § 4.2, we present the optimal strategies for generalization when the confounder is unobserved. In practice, we sometimes have access to proxies for the hidden confounding variable [4, 85, 62], which can be leveraged to substitute for the hidden confounding variable. We now study the impact of adding such non-causal but informative covariates on predictive information.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

Since $U \rightarrow Y$, any covariate that is informative to U , is also informative to Y and vice-versa. From the predictive information decomposition in (1), to maximize $I(Y; \hat{Y})$, it is required to minimize the

Table 2: Quantification of distribution shifts. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples, and statistical significance relative to a zero mean is evaluated with one-sample t-tests ($\alpha = 0.05$), confirming that all measures differ significantly from zero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] suggest based on their empirical analysis that collecting additional covariates $\mathbf{X}_I$ such that $\mathbb{P}(Y|\mathbf{X}, \mathbf{X}_I)$ is more stable across environments i.e., reducing concept shift, improves the OOD test accuracy. We theoretically show that utilizing more informative variables helps maximizing predictive information by maximizing or minimizing certain terms in (1).

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned}
 (i) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) &> I(\phi_1(\mathbf{X}); Y|E) & (ii) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) &> I(\phi_1(\mathbf{X}); E) \\
 (iii) \ I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) &< I(Y; E|\phi_1(\mathbf{X})) & (iv) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) &> I(\phi_1(\mathbf{X}); E | Y)
 \end{aligned} \tag{5}$$

Proposition 4.3 establishes that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii), while reducing concept shift (5.iii). At the same time, informative covariates also increase variation (5.iv), suggesting that their benefits may need to be accompanied by mechanisms that explicitly control this additional variation.

5 Experimental results

We evaluate the proposed account on both real-world and synthetic datasets in order to examine: (i) whether hidden confounding shift is detectable in real-world data, (ii) how the components of the decomposition in (1) shape OOD generalization under hidden confounding shifts, and (iii) how informative covariates contribute to improved generalization. Our real-world analysis uses eight tabular benchmark datasets—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—with datasets and domain splits drawn from the *TableShift* benchmark [24]. Model performance is assessed using ID and OOD test accuracies. We compare two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain-generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. Unless otherwise stated, we report averages over 5 random hyperparameter choices, as detailed in Appendix B; variation across random seeds is statistically insignificant for these datasets [24]. Additional experimental details are provided in Appendix B, and results with hyperparameter tuning appear in Appendix C.

Hidden confounding shift: Hidden confounding shifts induce observable changes in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. The measurements in Table 2 show that label shift, covariate shift, conditional covariate shift, and concept shift are all present in the real-world datasets, which is consistent with the presence of hidden confounding shifts. For a complementary qualitative analysis, we query GPT-4o [1] for potential hidden confounders in several benchmark datasets [24, 57]; these results are reported in Appendix E. In the *food stamps* dataset, for example, unmeasured state-specific *economic policies* may influence both *household income* and *food stamp reciprocity*.

Conditional informativeness vs. accuracy: From Proposition 4.2, under hidden confounding shift, maximizing the difference *conditional informativeness* – *residual* is essential for maximizing predictive information. Results on the first six datasets (that have more than one training environments [57, 24]) listed in Table 2 are shown in Figure 4. Dataset-wise results are in Appendix C. These results show that the difference *conditional informativeness* – *residual* is positively correlated with ID and OOD accuracy, with XGB being the best performing model with highest value of *conditional informativeness* – *residual*. As shown in the dataset-wise results in Appendix C, we

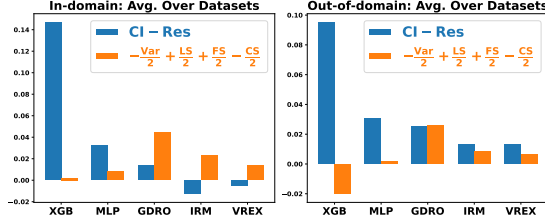


Figure 4: Across the plotted methods, the difference *conditional informativeness* – *residual* is positively associated with the average ID and OOD test accuracies over the eight datasets summarized in the table on the right.

Table 3: Across all methods, sign consistency is high for concept shift (CS); however, Proposition 4.2 identifies conditional informativeness (CI) as central to generalization, and XGB attains the strongest value on this metric. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric (↑)					ID-Test Accuracy (↑)			OOD-Test Accuracy (↑)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

observe that in many settings, due to hidden confounding shifts, the bars denoted with orange are close to zero because the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ is zero under hidden confounding shift (Proposition 4.2). The relative ID / OOD performance of the methods shown in Figure 1 is consistent with previously reported results in the literature [57, 24].

Informative covariates vs. accuracy: We now study how inclusion of informative covariates helps in achieving better OOD generalization. To this end, we adopt the covariate partitioning from [57], where all available covariates are grouped into three nested covariate subsets: *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (see Appendix § B). We evaluate how the terms in the decomposition in (1) affect ID and OOD performance when going from one covariate subset to another covariate subset. To this end, we use what we call the *sign consistency metric*, which works as follows. When we move from one covariate set to a larger covariate set, for any term in (1) with a positive coefficient, such as conditional informativeness, we count how often its value rises when accuracy improves. For negatively weighted terms, such as concept shift, we count how often its value decreases when accuracy improves. The metric is the fraction of observations where a term’s change aligns with its *beneficial* direction, thus capturing how reliably it contributes to better generalization. See Appendix § B for the formal definition. Table 3 shows that both conditional informativeness and concept shift exhibit high sign consistency with accuracy gains. While reducing concept shift is crucial, XGB further enhances conditional informativeness compared to other methods, yielding additional performance improvements. Dataset-specific results are presented in Appendix § C.

In real-world datasets, verifying whether *all available covariates* contain *every relevant or sufficiently informative* variable is often infeasible. We therefore further test the hypothesis that informative covariates enhance generalization using synthetic data with known causal structure: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes informative covariates caused by the hidden confounder U . As Figure 5 shows, including $\mathbf{X}_I$ increases conditional informativeness and feature shift while reducing concept shift. The variation component remains identically zero because of the simplicity of the problem. The synthetic data-generating process and additional results are reported in Appendix § D.

6 Discussion, limitations, and future work

In summary, we decompose the predictive information shared by model outputs and true outcomes in order to expose the factors that constrain OOD generalization (Equation (1)). We argue that hidden confounding shifts, manifested through environment-specific changes in covariate and outcome distributions, are difficult to avoid in realistic settings and that no finite collection of observed covariates or proxy variables can completely eliminate the ambiguity they introduce. Because confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), strategies

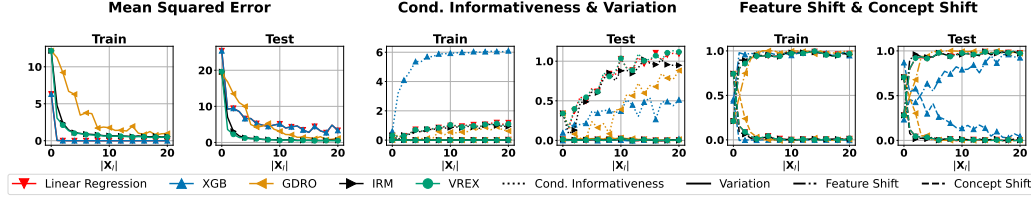


Figure 5: Adding proxy variables $\mathbf{X}_I$ for U that are informative about Y reduces MSE, increases conditional informativeness and feature shift, and decreases concept shift.

that treat these shifts as separable phenomena are inherently limited. Our analysis shows that learning environment-specific input–output mappings can be more effective than imposing a single invariant predictor (Equation (4)). By foregrounding the inevitability of hidden confounding and the necessity of addressing it directly, this work provides a foundation for more robust OOD methods. Important open challenges include: (i) developing algorithms that remain effective under limited confounder overlap, (ii) quantifying the cost–accuracy trade-off associated with acquiring informative covariates, (iii) handling entangled shifts without relying on untestable proxy assumptions, and (iv) modeling ambiguity from unobserved confounders in ways that support new OOD-robust paradigms.

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687 Appendix

688 A Proofs of propositions

689 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 690 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 691 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

692 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 693 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

694 *Proof.* In the causal graph in Figure 3 of the main paper, the substructure $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$
 695 implies that $\hat{Y}$ contributes no additional information for predicting Y once either $\phi(\mathbf{X})$ or $\mathbf{X}$ is
 696 known. The path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$ between $\hat{Y}$ and Y is blocked by conditioning on either
 697 $\{\phi(\mathbf{X})\}$ or $\{\mathbf{X}\}$, yielding $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. We therefore expand $I(Y; \hat{Y})$ as follows.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

698 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

699 Summing Equations (6) and (7), and then collecting like terms, yields the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

700 The analogous decomposition with $\mathbf{X}$ in place of $\phi(\mathbf{X})$ follows by using $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$ and introducing
 701 $\mathbf{X}$, rather than $\phi(\mathbf{X})$, into the expansion. $\square$

702 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 703 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 704 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

705 *Proof.* Using the causal graph in Figure 3 of the main paper, we prove the proposition by treating the
 706 two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately.

707 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 708 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

709 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 710 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 711 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

712 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

713 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

714 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

715 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 716 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 717 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

718 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 719 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

720 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

721 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

722 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

723 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 724 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

725 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 726 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

727 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
728 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
729 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
730 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

731 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
732 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
733 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
734 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
735 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

736 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

737 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

738 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
739 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
740 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
741 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
742 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

743 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
744 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
745 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
746 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
747 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

748 B Experimental setup

749 This section specifies the experimental setup. Because random seeds have been shown to have
750 statistically insignificant effects in many of these settings [24], the main-paper results report the
751 mean and standard deviation over five randomly selected hyperparameter values, sampled from the
752 corresponding domains in Table B1, for each dataset–method combination. Since performance is
753 evaluated on both ID and OOD test data, mutual-information quantities are estimated using the
754 nonparametric entropy-estimation toolbox [37, 77, 78]. Following [24], we compute these quantities
755 on in-distribution (ID) test data from training domains and out-of-distribution (OOD) test data from
756 test domains.

757 For consistency, we draw 20,000 random samples
758 from each ID and OOD test set whenever
759 possible, although several datasets contain fewer
760 than 20,000 observations. Across experiments,
761 sample size has only a limited effect on the es-
762 timated mutual-information terms. For example,
763 Figure B1 shows that the terms in Equation (1),
764 computed from XGB representations on the *in-*
765 *come* dataset, remain relatively stable as sample
766 size varies. Features $\phi(\mathbf{X})$ are extracted from
767 the layer preceding the classification head; for
768 XGBoost, we combine model output margins
769 with SHAP values. All experiments were conducted on a single NVIDIA RTX A6000 GPU.

770 In our experiments studying the impact of informative covariates on predictive information, we follow
771 the categorization introduced by Nastl and Hardt [57], partitioning covariates into: *causal*, *arguably*
772 *causal*, and *all covariates*. Causal covariates have clearly established influence on the target, with well-
773 supported directionality and unlikely reverse causation. However, strict reliance on causal variables
774 risks omitting relevant parents due to knowledge gaps. Arguably causal covariates have uncertain
775 causal relationships, meeting at least one of: (1) being known causal, (2) having plausible but poten-
776 tially bidirectional influence, or (3) likely (but unconfirmed) causal effect. These groups approximate

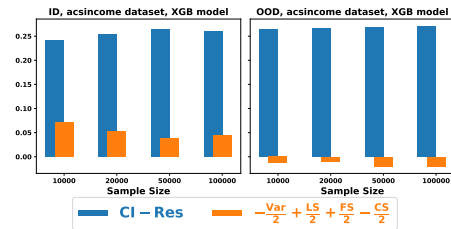


Figure B1: Estimated predictive-information terms remain comparatively stable across sample sizes on the *income* dataset.

Table B1: Hyperparameter search spaces considered for each model in the experimental evaluation.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

777 true causal parents based on available knowledge, though relationships may be confounded. Some
778 datasets additionally contain *anti-causal* covariates where the target likely affects them but not vice
779 versa. The *all covariates* set includes all observed variables regardless of causal status. The sign con-
780 sistency metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
781 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
782 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

783 Here, P denotes the set of covariate-set pairs (s, t) such that $t \supset s$; for example, s may be the
784 causal-covariate set and t the arguably causal-covariate set. The factor of 2 in $2P$ accounts for the

785 use of both training and test data. $\mu_s^{\text{tr/te}}(m)$ denotes the mean of measure m for covariate set s in
786 training or test data, and $\mathbb{I}(\cdot)$ is the indicator function. For the covariate settings $\{\text{causal}, \text{arguably}$
787 $\text{causal}, \text{all}\}$, $|P| = 3$. The final sign-consistency value for each m is obtained by averaging across
788 datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, where $|D|$ is the number of datasets.

789 C Additional results on real-world datasets

790 This section reports additional real-world dataset results. Figures C2 and C3 compare, for each
791 dataset, the terms *conditional informativeness – residual* and *–variation / 2 + label shift / 2 +*
792 *feature shift / 2 – concept shift / 2*, whose aggregate behavior appears in the left panel of Figure 4
793 in the main paper. The central pattern is that, as implied by Proposition 4.1, the sum
794 of the blue and orange terms is positively correlated with overall model performance. At the
795 same time, because real-world data may exhibit hidden confounding shift, the contribution of
796 *conditional informativeness – residual* to overall predictive information is larger than the contri-
797 bution of *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*. This pattern further
798 motivates maximizing conditional informativeness when seeking stronger OOD generalization under
799 hidden confounding shift.

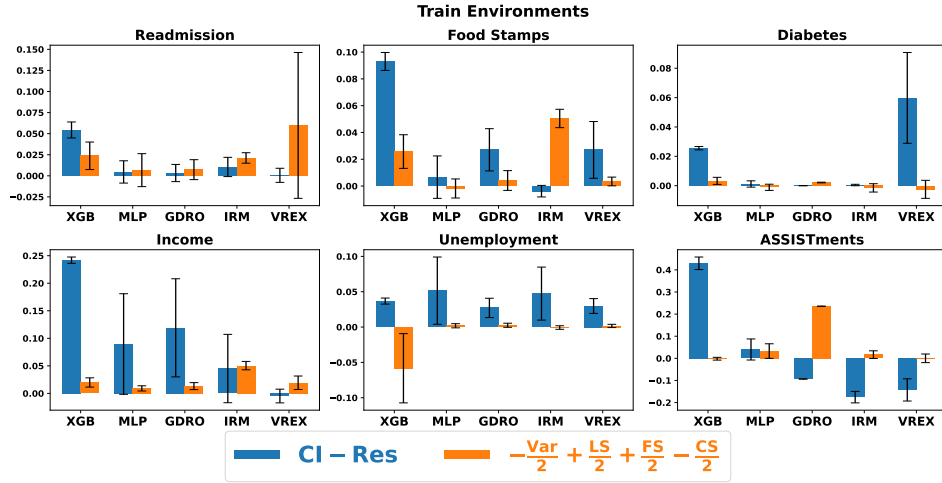


Figure C2: Dataset-specific decomposition results evaluated on ID data.

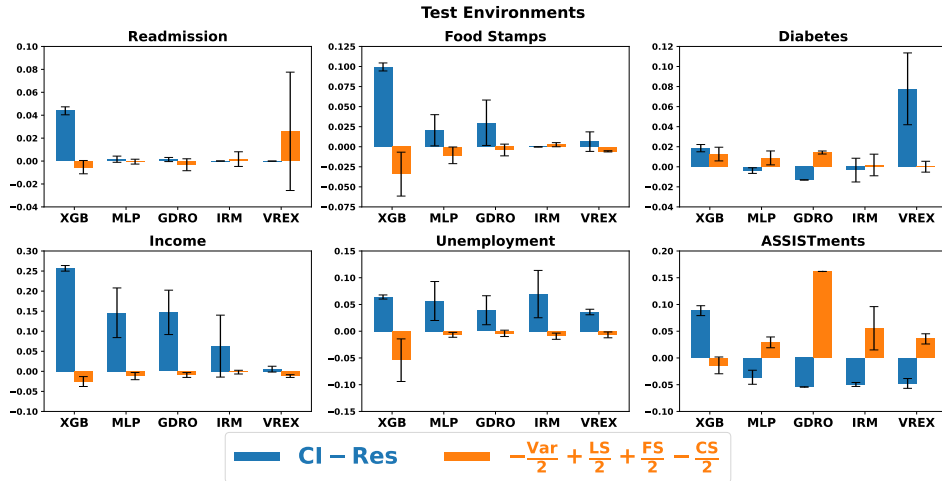


Figure C3: Dataset-specific decomposition results evaluated on OOD data.

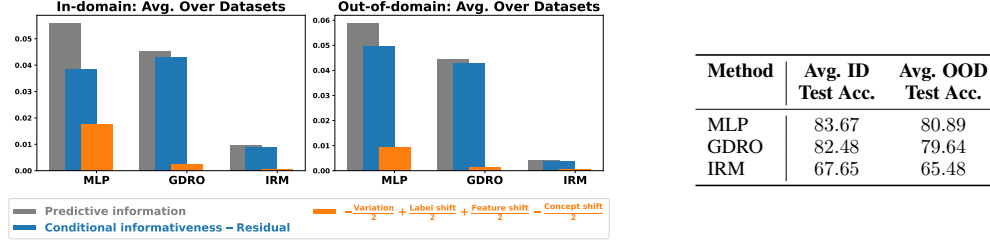


Figure C4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

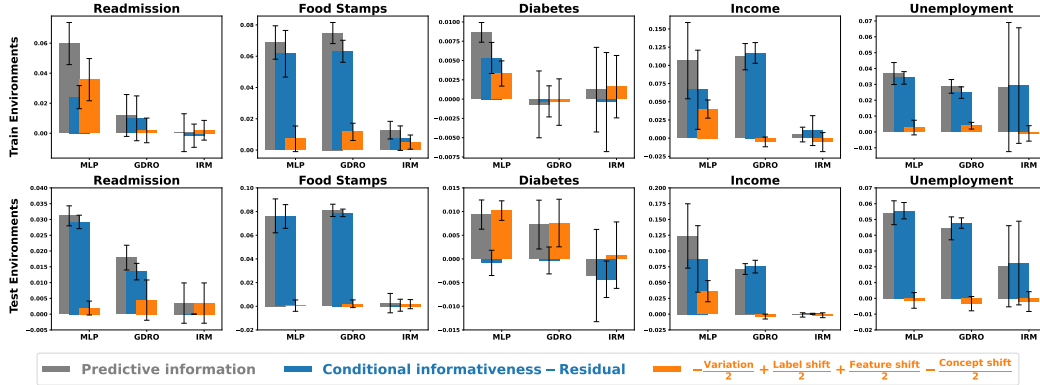


Figure C5: Dataset-specific comparison of the terms contributing to predictive information.

Table C2: Comparison of model ID and OOD test accuracies.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further show the comparison of various terms of the information decomposition in (1) for each method and dataset. From these results, we observe that the values plotted in the second column (*variation*) and third column (*feature shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. Similarly, we observe that the values plotted in the fourth column (*label shift*) and fifth column (*concept shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. In almost all settings, the difference *conditional informativeness* - *residual* is the majority contributor for the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run

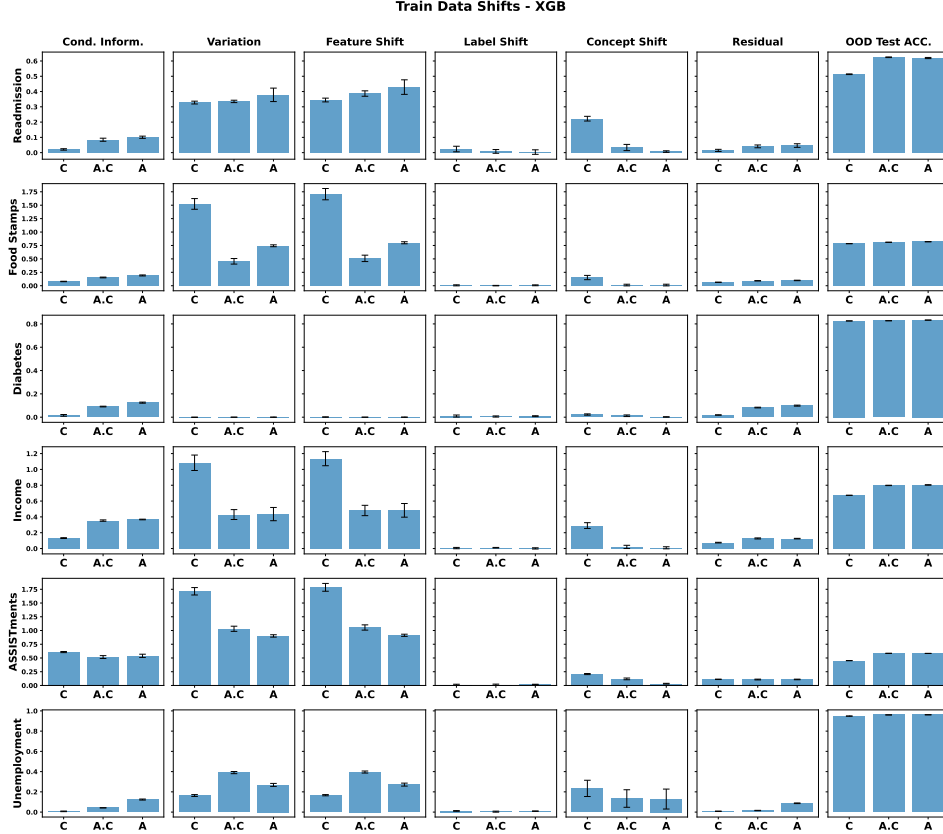


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. Dataset-specific results shown in Figure C5 and Table C2 also show similar trends.

Potential limitations with predictive information decomposition analysis: While predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, it does not guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while maintaining interpretability.

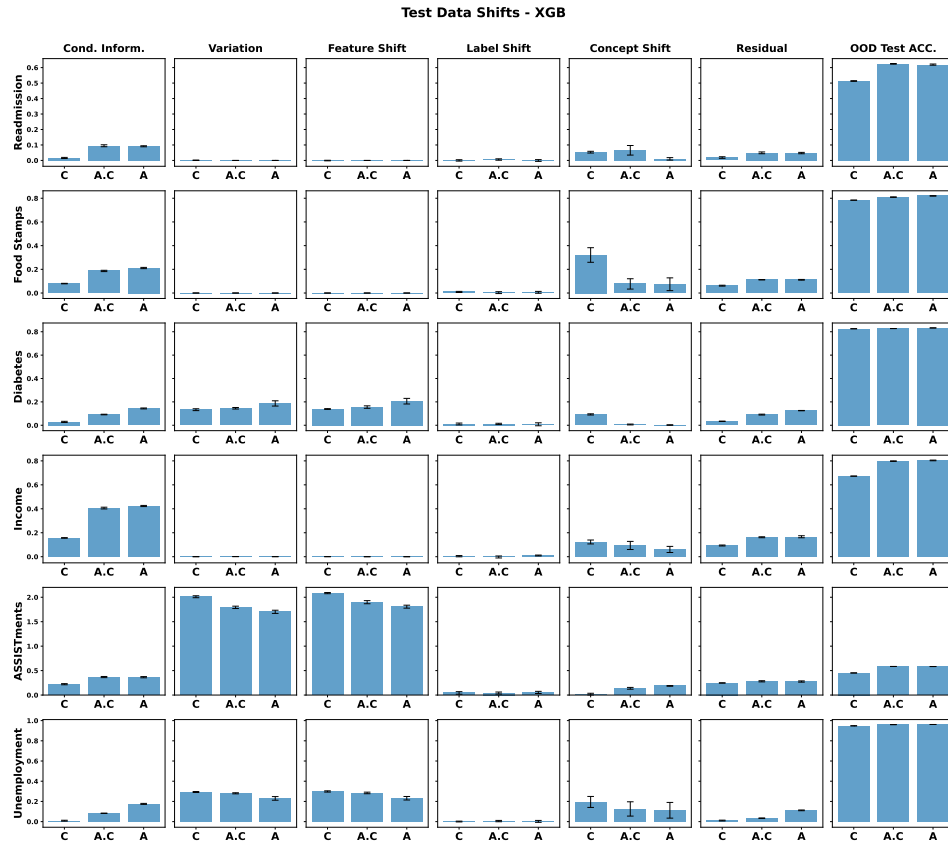


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

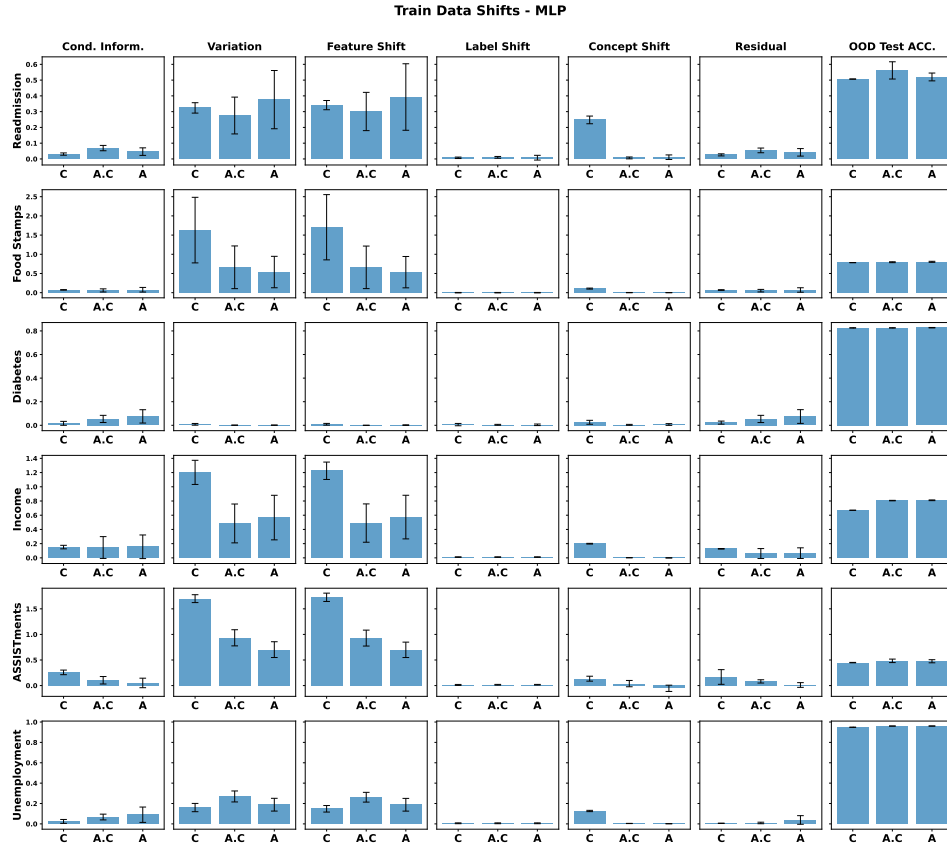


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

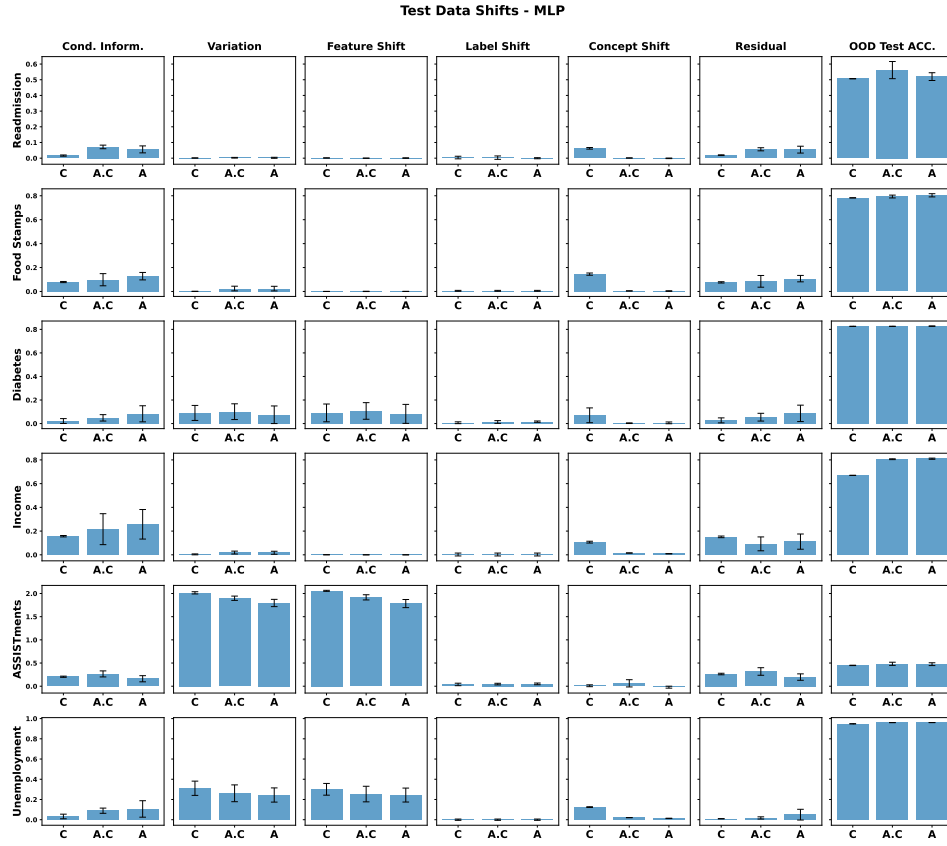


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

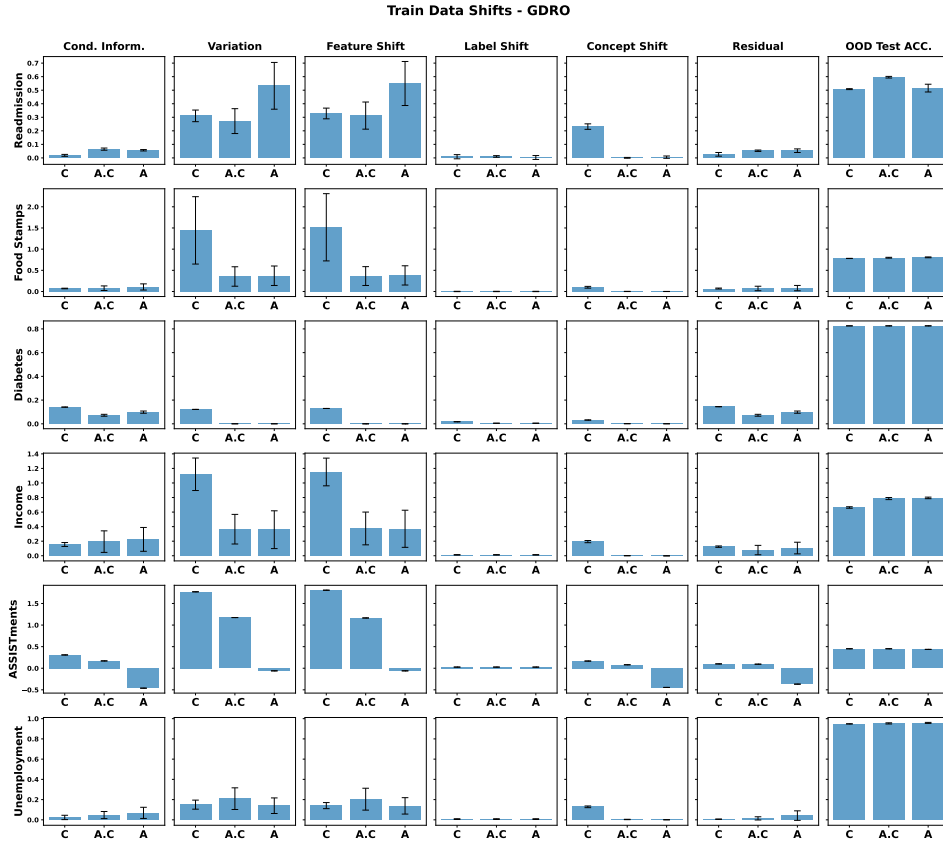


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

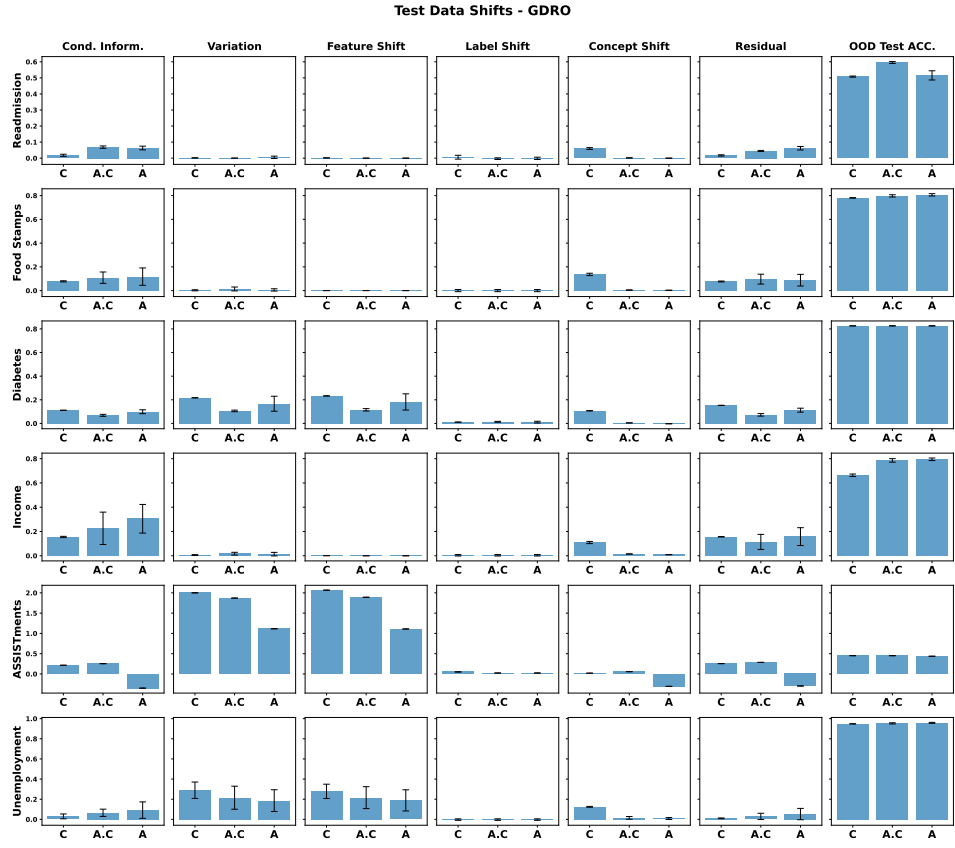


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

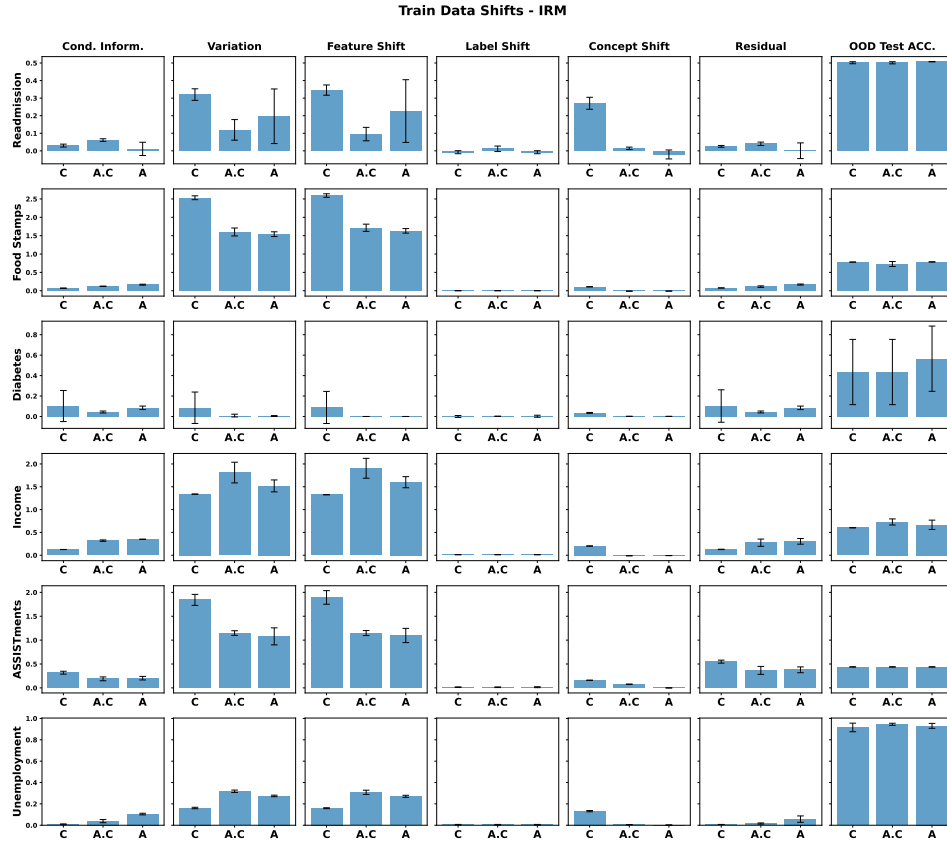


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

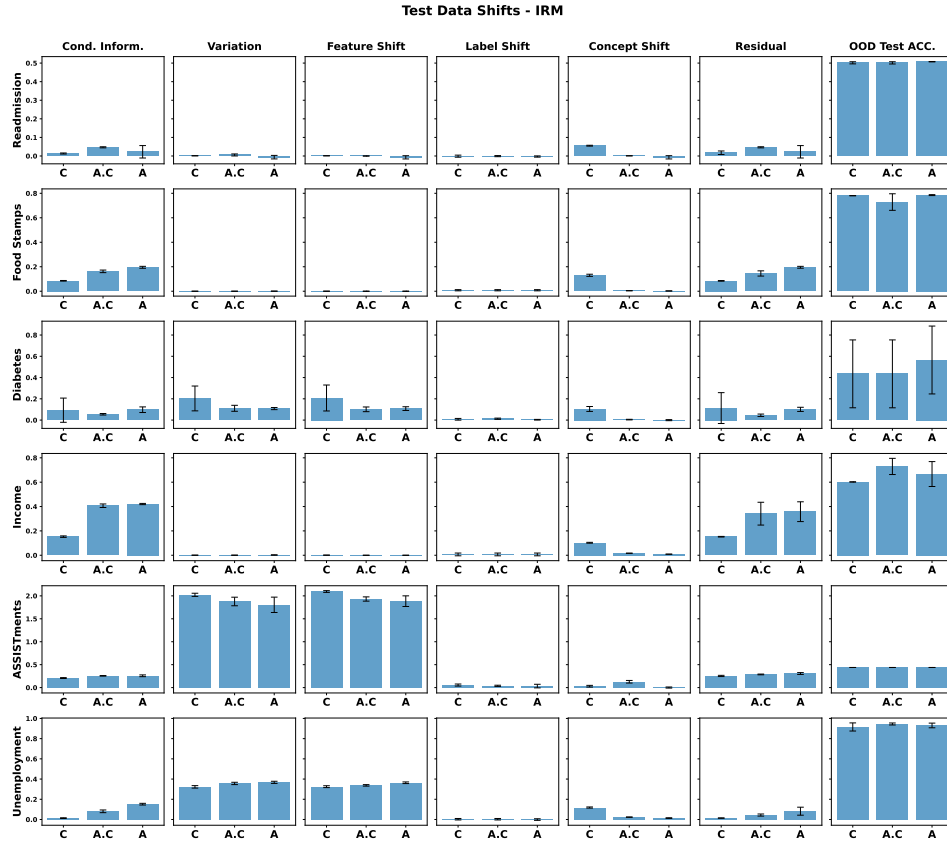


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

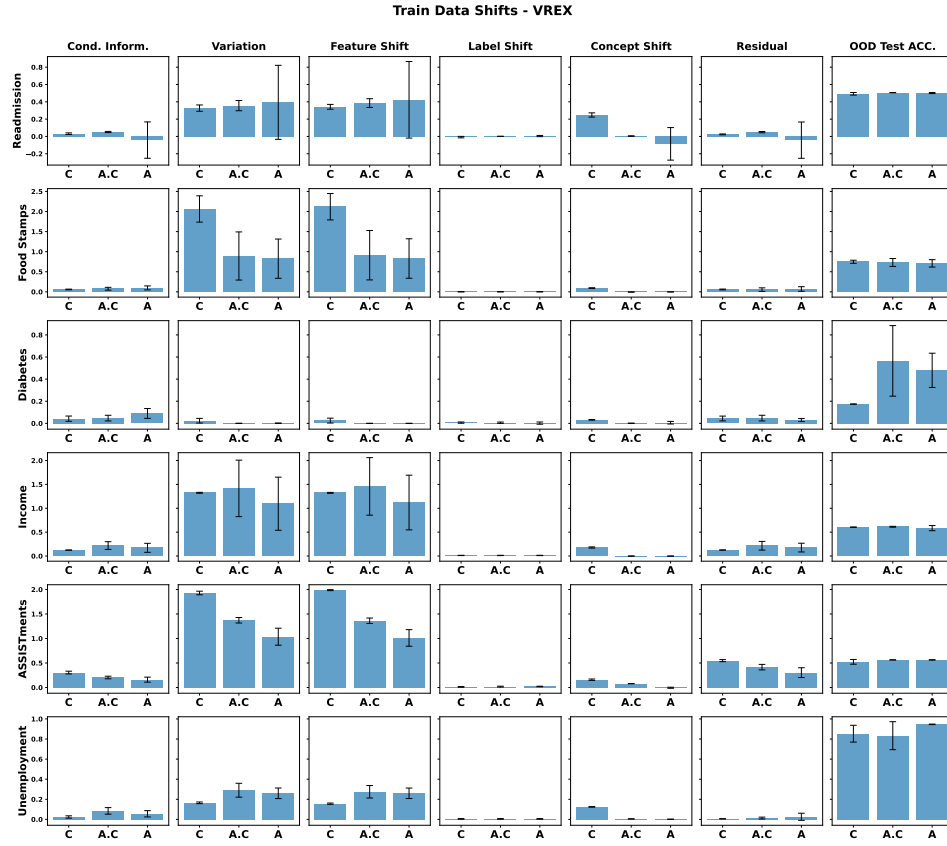


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

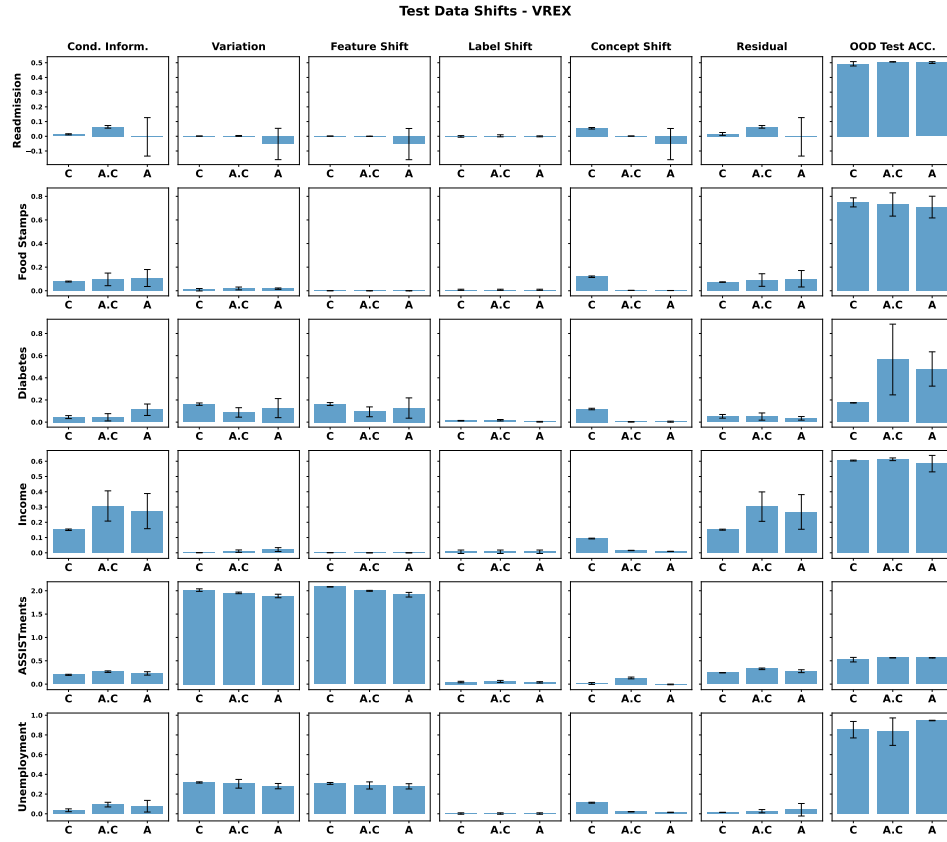


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

831 D Results on synthetic datasets

832 We conduct several synthetic-data experiments to examine: (i) the effects of alternative informative-
833 ness criteria, (ii) how confounder-overlap levels across environments influence model performance,
834 and (iii) how the ratio of causal to anti-causal variables affects performance. The data-generating
835 process follows the structural equations in (12) and instantiates the causal structure $U \rightarrow X, U \rightarrow Y,$
836 $U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is an unobserved confounder, X is an observed covariate, Y is the
837 target variable, and $\mathbf{X}_I$ denotes additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

838 We consider multiple functional forms for f_X , f_i , and f_Y , with the corresponding results reported in
839 the subsequent analysis. Domain shifts are induced by systematically varying the environment-specific
840 confounder mean μ_u^e , thereby allowing us to evaluate how different magnitudes of distributional shift
841 affect model performance across environments.

842 **Informativeness vs. accuracy:** We first use linear structural equations, $f_X = 0.3U$, $f_i = 0.1U$, and
843 $f_Y = X - 2U$, with $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test environments.
844 For the results in Figure 5 of the main paper, we consider one hidden confounder and $|\mathbf{X}_I| = 20$
845 informative covariates. Increasing the number of informative covariates from 0 to 20 produces
846 (i) lower mean squared error, (ii) greater conditional informativeness, (iii) larger feature shift, and
847 (iv) lower concept shift. We extend this finding through two additional informativeness criteria.
848 First, with one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$), we vary noise
849 levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$; Figure D16 shows that decreasing noise from 2.0 to 0.0 again reduces
850 mean squared error, increases conditional informativeness and feature shift, and decreases concept
851 shift. As in the other synthetic experiments, the variation term remains zero throughout. Finally,
852 we study a setting with 20 hidden confounders and 20 corresponding informative covariates, using
853 modified structural equations $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17 shows that
854 introducing informative covariates associated with hidden confounders produces the same qualitative
855 pattern: reduced mean squared error, improved conditional informativeness, enhanced feature shift,
856 and decreased concept shift. These experiments demonstrate that multiple operationalizations of
857 informativeness can facilitate OOD generalization.

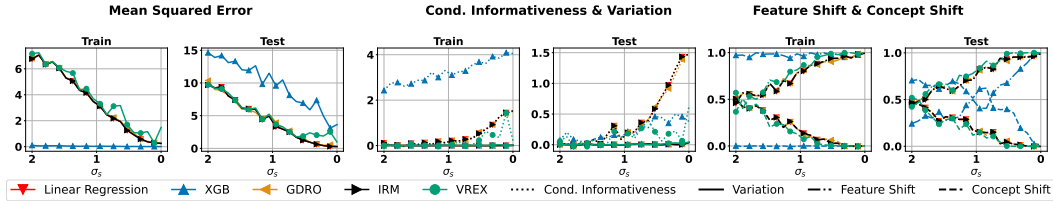


Figure D16: Adding a low-noise proxy variable reduces MSE, increases conditional informativeness and feature shift, and decreases concept shift.

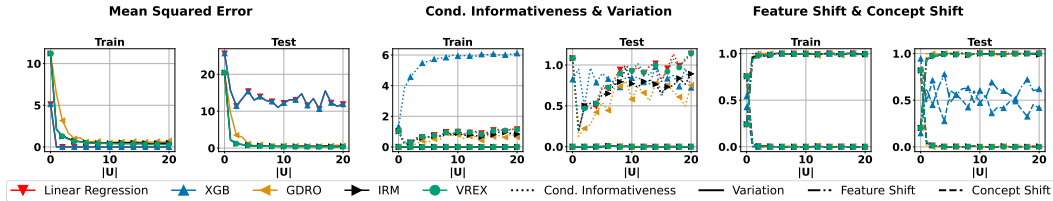


Figure D17: Adding proxy variables $\mathbf{X}_I$ for hidden confounders $\mathbf{U}$ that are informative about Y reduces MSE, increases conditional informativeness and feature shift, and decreases concept shift.

858 **Overlapping confounder support vs. accuracy:** As established in the main paper, recent work
859 by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that
860 assumes test confounder support remains within training support. While our results demonstrated

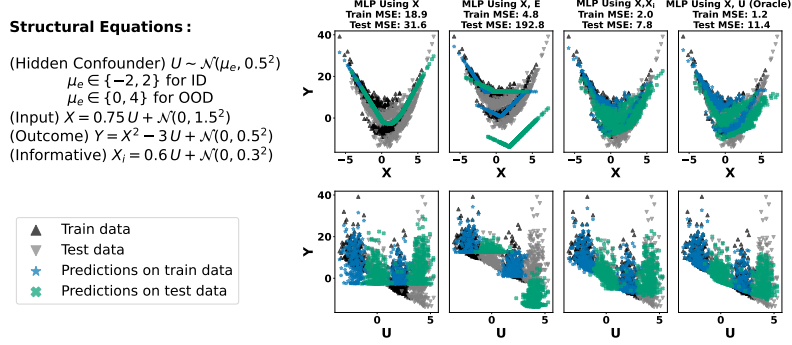


Figure D18: MLP performance in the low-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

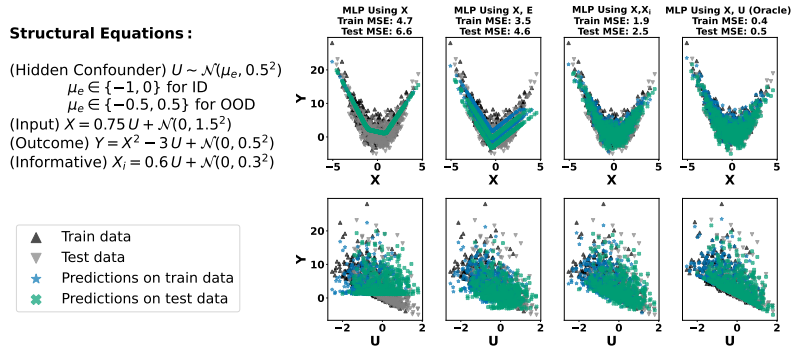


Figure D19: MLP performance in the high-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

861 successful learning of correct relationships in synthetic linear data when sufficient information about
862 hidden confounders exists in observed covariates, we now extend this analysis to nonlinear data
863 using MLP and XGB models under both low and high confounding support conditions (Figures D18–
864 D21). In low confounding overlap settings (characterized by distant μ_e between ID and OOD data),
865 models relying solely on X learn an inadequate global function that fails to distinguish environments.
866 Performance improves when incorporating environment-specific summary statistics of observed
867 covariates, which helps capture environment-specific relationships, but the most significant gains
868 occur only when leveraging additional informative covariates, underscoring their importance. For
869 completeness, we include both oracle model results and U - Y relationship scatter plots to elucidate
870 the learned input-output mappings. These experiments confirm better performance under high
871 confounding overlap as expected, but more importantly, reveal promising results in the low-overlap
872 regime - a previously unstudied scenario that challenges the common support assumption in the
873 literature.

874 **Number of causal vs anti-causal variables:** Recent benchmarks [24] show that baseline methods
875 (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO,
876 IRM, VREX) in real-world datasets, coinciding with the prevalent causal structure $\mathbf{X} \rightarrow Y$ in
877 tabular data. To test if baseline superiority stems from the causal structure: $\mathbf{X} \rightarrow Y$, we vary
878 the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup$
879 $\mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). Results (Figure D22) show baseline resilience
880 even under the causal structures $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, though no method dominates universally
881 (Figures D23, D24). Baselines achieve superior PMA-OOD scores [24] (fraction of maximum
882 OOD accuracy). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP,
883 ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD
884 generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43],
885 DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods
886 (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

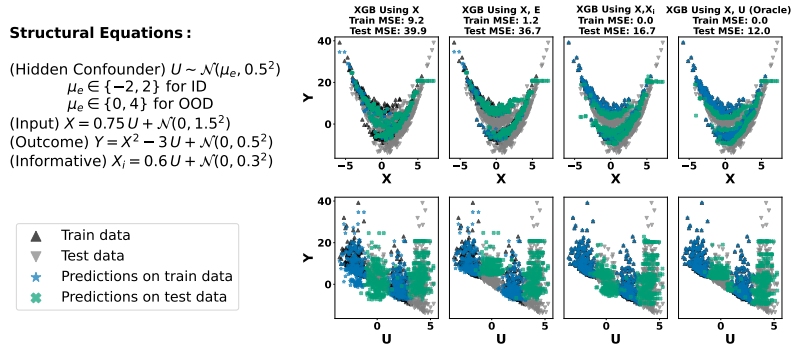


Figure D20: XGB performance in the low-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

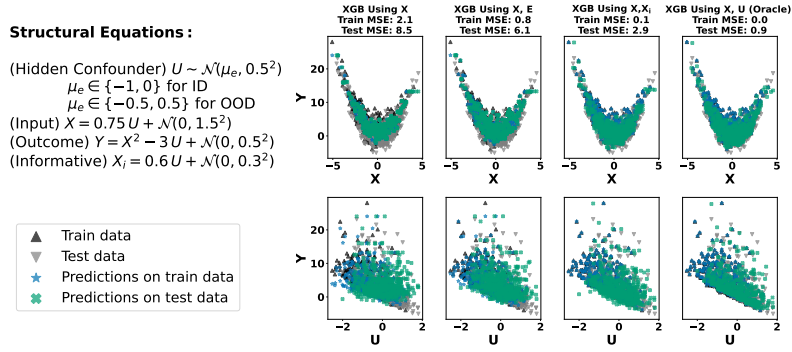


Figure D21: XGB performance in the low-overlap non-linear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

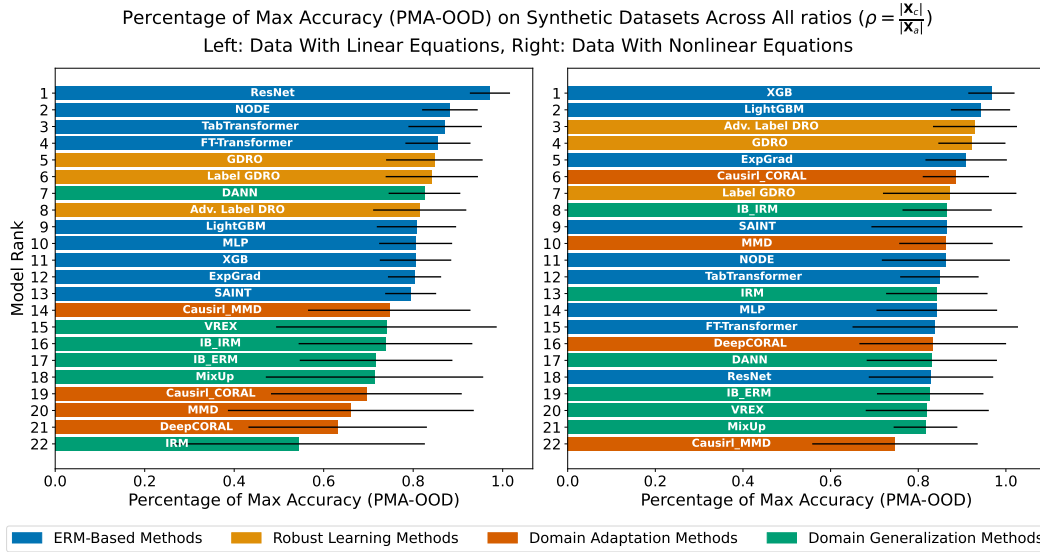


Figure D22: ERM-based methods attain higher percentage-of-maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, consistent with the real-world findings of [24].

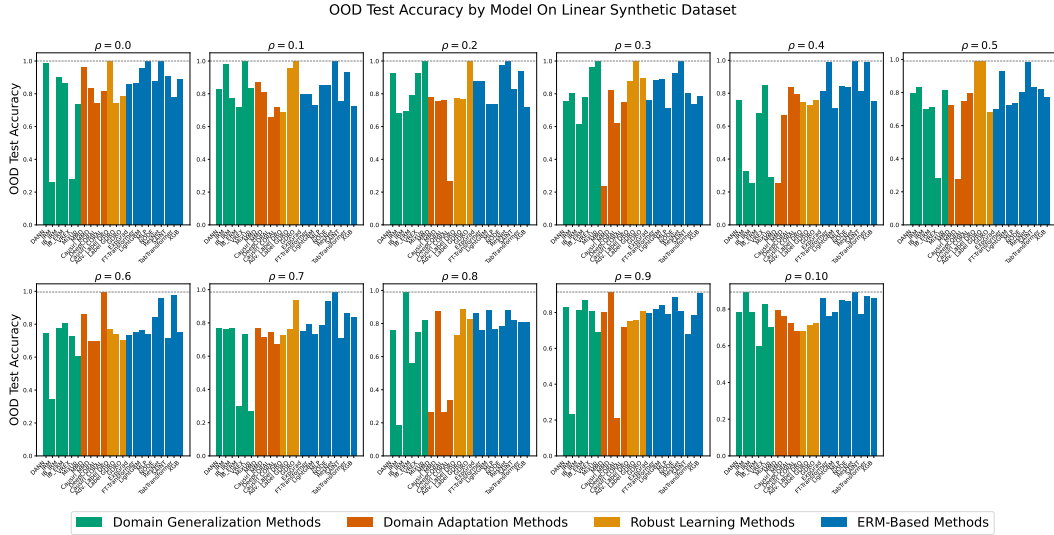


Figure D23: Model-specific OOD test accuracy in the linear setting across different values of ρ .

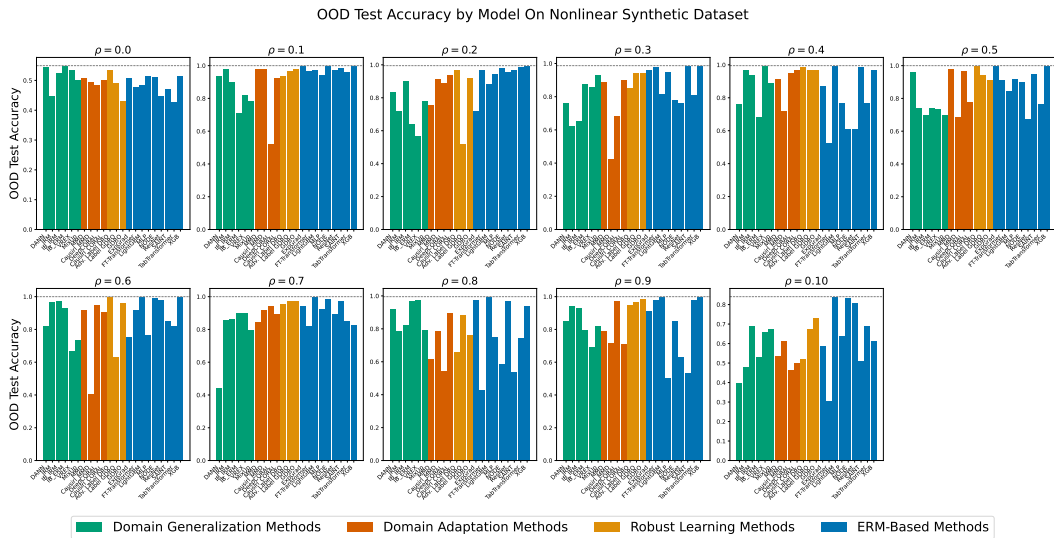


Figure D24: Model-specific OOD test accuracy in the non-linear setting across different values of ρ .

E Qualitative analysis of hidden confounding shift in real-world datasets

This section provides a qualitative assessment of hidden confounding shifts in real-world datasets. Tables E3-E18 enumerate potential hidden confounders that may influence observed covariates across several real-world datasets [57]; additional dataset details are available in [24, 57]. We use GPT4o [1] for this elicitation task and emphasize that the resulting lists of unobserved confounders are illustrative rather than exhaustive. The qualitative analysis supports the broader argument that hidden confounding shifts plausibly arise across domains in real-world data and can materially affect OOD generalization.

Prompt: We use the following prompt to query GPT4o for candidate hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.